



Experimental and kinetic modeling study of laminar burning velocities of NH₃/syngas/air premixed flames

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ABSTRACT

Ammonia (NH₃) can be used as carbon-free alternative fuel for modern energy and transportation systems. Co-firing NH₃ with syngas can overcome the high ignition energy and low burning velocities of pure NH₃ flames on the one hand, while regarding the characteristics of syngas on the other hand, this strategy may have low-emission potential in real application, and a corresponding research can be helpful for validating or developing NH₃ co-firing mechanisms with more complex fuels. The present study experimentally investigated laminar burning velocities of NH₃/syngas/air flames at atmospheric pressure and 298 K using the heat flux method. Two types of syngas components were used, i.e., SYN_A: 5 vol% H₂ + 95 vol% CO and SYN_B: 50 vol% H₂ + 50 vol% CO, and the measured conditions cover wide ranges of mixing ratios and equivalence ratios. Several literature kinetic mechanisms were tested and a new mechanism was proposed. Results calculated by the present mechanism agree well with experimental data of the burning velocities and the ignition delay times of NH₃, NH₃/H₂, NH₃/CO, and NH₃/syngas flames at various mixing ratios, equivalence ratios, and pressures. The present mechanism also reproduces the trend of NO_x emission characteristic in literature. Detailed kinetic analyses using the present mechanism were carried out, showing the NH₃ oxidation processes in NH₃/syngas/air flames and the most rate-limiting reactions for predicting the laminar burning velocities. Important reactions with different rate parameters from different sources were labeled, which could be helpful for future organization or optimization of NH₃ kinetic mechanisms.

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1. Introduction

Ammonia (NH₃) has caught more and more attention currently in the combustion community for its carbon-free property as possible alternative fuel for modern energy and transportation systems. With high boiling temperature (−33.4 °C at 1 atm) and high volumetric energy density (12.7 MJ/L at −33.4 °C and 1 atm, calculated by Lower Heating Value), ammonia has more advantages in storage and transportation compared with hydrogen (H₂) [1–4]. Ammonia can be produced from H₂ and N₂ through well-established Haber–Bosch process and has been utilized in fertilizer industry for hundred years. Besides fertilizer industry, ammonia is also widely used in refrigerant industry and deNO_x process in power industry by selective catalytic reduction (SCR) and selective non-catalytic reduction (SNCR). With established ammonia storage and distribution infrastructure, ammonia can be a good candidate to replace fossil fuels with less investment in near future [4]. The combustion of NH₃ has been tested against engines, boilers and

gas turbines [5–7], and strategy of co-firing NH₃ with other fuels have been adopted facing the high ignition energy and low burning velocities of pure NH₃ flames [8,9]. These co-firing fuels include H₂ [10], methane (CH₄) [5], dimethyl ether (DME) [11], gasoline [12], and diesel [6] etc., while there are other more fuels that have yet been tested to co-fire with NH₃, among which, syngas can also be a good option.

Syngas comprises of H₂ and carbon-monoxide (CO) as main components, with CO₂, H₂O, and other impurity compounds as diluted species. In industrial processes like Integrated Gasification Combined Cycle (IGCC), syngas has been found to have high energy conversion ratio and low emissions compared with direct combustion of coal [13]. More fundamentally, the knowledge of syngas oxidation is essential in the development of combustion mechanisms for larger fuel molecules [14]. Regarding the above mentioned characteristics of NH₃ and syngas, their co-firing may have good combustion property and low-emission potential in real application, and a corresponding research can be helpful for validating or developing NH₃ co-firing mechanisms with more complex fuels.

Fundamental experiments of NH₃ and syngas co-firing combustion are scarce, and it is found existing works that include NH₃ and syngas at the same time treat the NH₃ as impurity species in

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Table 1

Cases measured by the present experiments.

Fuel type	Description	S_L data points
NH ₃ +SYN_A	$\phi = 1.0$, $x_{\text{SYN}_A} = 0-1.0$	42
	$x_{\text{SYN}_A} = 0.2$, $\phi = 0.85-1.4$	12
	$x_{\text{SYN}_A} = 0.4$, $\phi = 0.7-1.6$	19
	$x_{\text{SYN}_A} = 0.6$, $\phi = 0.7-1.6$	19
NH ₃ +SYN_B	$\phi = 1.0$, $x_{\text{SYN}_B} = 0-0.7$	29
	$x_{\text{SYN}_B} = 0.2$, $\phi = 0.8-1.4$	13
	$x_{\text{SYN}_B} = 0.4$, $\phi = 0.7-1.6$	19
	$x_{\text{SYN}_B} = 0.6$, $\phi = 0.7-1.6$	19

Table 2

Literature kinetic mechanisms used in this work.

Mechanism	Year	Main target
Dagaut et al. [34]	2008	NO formation
Klippenstein et al. [35]	2011	NO formation
Mathieu and Petersen [23]	2015	NH ₃ ignition
Song et al. [30]	2016	High-pressure NH ₃ oxidation
Nakamura et al. [31]	2017	Low-temperature NH ₃ /air ignition
Zhang et al. [26]	2017	H ₂ /NO _x and syngas/NO _x systems
San Diego [25]	see text	NH ₃ ignition
Otomo et al. [32]	2018	NH ₃ and NH ₃ /H ₂ combustion
Okafor et al. [24]	2018	NH ₃ /CH ₄ /air system
Shrestha et al. [27]	2018	NH ₃ and NH ₃ /H ₂ system
Mei et al. [33]	2019	oxygen enriched NH ₃ system

syngas rather than a main component in the fuel mixture. For example, the composition of one mixture in the ignition delay time study by Mathieu et al. [15] is 0.0002 NH₃/0.5 H₂/0.5 CO/0.01 O₂/0.9798 Ar (mole basis); and in the flame stability and NO_x formation study by Lee and Kil [16], the NH₃ concentration in the unburnt mixture is not larger than 3000 ppm. The far less NH₃ concentration than syngas in these mixtures can make the tested combustion process mostly governed by syngas chemistry, thus more experimental work with higher NH₃ concentration is needed to validate current chemical mechanisms used for NH₃/syngas co-firing conditions.

There are several numerical mechanisms available that include NH₃ and syngas as main components at the same time, e.g., Tian et al. mechanism [17], Mendiara and Glarborg mechanism [18]. However, results calculated by some of these mechanisms have been found deviating from experimental burning velocity data of NH₃/H₂/air and NH₃/CO/air flames in our latest publication [19], thus, the performance of these mechanisms for NH₃/syngas co-firing combustion modeling needs to be carefully validated.

Therefore, this study especially focuses on the combustion of NH₃/syngas. Experiments were carried out using the well-developed heat flux method to obtain adiabatic laminar burning velocities of NH₃/syngas/air flames with different mixing ratios and equivalence ratios. The experimental data was compared against the predicted laminar burning velocities using several literature mechanisms. A self-designed new mechanism was also developed with NH₃/syngas as target. The new mechanism was validated against wide range of experimental data for laminar burning velocities, ignition delay times, and NO_x emission characteristics. Detailed kinetic analyses for the NH₃ oxidation process were also discussed.

2. Experiment and kinetic modeling

2.1. Experimental details

The experiment setup used in the present work was the same as that used in our previous publication [19], and thus only the most important points are shown here. The heat flux burner was used to measure the laminar burning velocities (S_L) of fuel+air mixtures at atmospheric pressure and 298 K unburnt temperature. The fuels investigated here were NH₃ mixed with two types of syngas, i.e., SYN_A: 5vol% H₂ + 95 vol% CO, and SYN_B: 50 vol% H₂ + 50 vol% CO. In the following, the unburnt fuel mixture is referred to by the mole fraction of syngas in the total fuel, e.g., $x_{\text{SYN}_A} = 0.5$ denotes that there are 50 vol% syngas and 50 vol% NH₃ in the fuel, and the syngas comprises of 5 vol% H₂ and 95 vol% CO.

Table 1 shows all the fuel mixtures investigated in the present experiments. For each fuel type, the stoichiometric mixtures were first measured at varying syngas mixing ratios; then at $x_{\text{SYN}} = 0.2$, 0.4, and 0.6, respectively, the dependence of S_L on the equivalence ratio ϕ was also tested. The initial mole fractions of each reactant

are also shown in Supplementary material with all the measured results.

All gases used in the present experiment had purities higher than 99.99% (Jingong Gas, Hangzhou, China). Same as discussed in [19], flame buoyancy and NH₃ adsorption on the burner were considered insignificant for the S_L measurement. The uncertainty of measured S_L were evaluated from the temperature readout scatter of the thermal couples beneath the burner plate, and the uncertainty of the gas flow rates controlled by several mass flow controllers (Alicat, 21-1-08-1 and 21S-1-32-1), both of which have been explained in detail elsewhere [20,21]. It was found that the evaluated value of S_L uncertainty was less than ± 0.8 cm/s for all the cases measured in the present work, and the S_L uncertainty will be shown as error bars attached to the experiment data.

2.2. Kinetic modeling details

All numerical simulations of S_L and burner stabilized flame in the present study were performed using CHEMKIN-II [22], Pre-mixed module. For each case studied, at least 200 grid points were used with CRUV 0.01 and GRAD 0.01. Thermal diffusion (Soret effect) and multicomponent transportation were taken into consideration, without which, the calculated S_L value was found to deviate as much as 1–2 cm/s for some cases investigated here.

Simulations of ignition delay times were performed using CHEMKIN-II [22], Closed Homogeneous module. The literature shock tube experimental data used for comparison were taken from Mathieu and Petersen [23] and Mathieu et al. [15], both of which used chemiluminescence emission from $A^2\Sigma^+ \rightarrow X^2\Pi$ transition of the excited-state hydroxyl radical (OH⁺) to identify ignition. With same strategy suggested by Mathieu and Petersen [23] and Mathieu et al. [15], the intersection of lines drawn along the steepest rate-of-change of OH⁺ profile and a horizontal line defining the zero-concentration level was used as ignition time. The small pressure scatterings reported by Mathieu and Petersen [23] and Mathieu et al. [15] have also been taken into account in the calculation processes.

There will be four mechanisms used in Section 3.1 to compare directly with the present S_L experimental data, namely, Okafor et al. mechanism [24], San Diego mechanism [25], Zhang et al. mechanism [26], and Shrestha et al. mechanism [27]. The San Diego mechanism [25] used herein refers to combination of the nitrogen chemistry (2018/07/23) and the main mechanism (2016/12/14). Both Okafor et al. mechanism and San Diego mechanism have been used in our previous work [19], showing better agreement with experimental results of NH₃/H₂/air and NH₃/CO/air flames than GRI-mech 3.0 [28].

In Section 3.2, after introducing the present mechanism targeting at the NH₃/syngas flames, other seven different mechanisms are used to make comparison of S_L , ignition delay times, and NO_x emission characteristics, which are all listed in Table 2 together

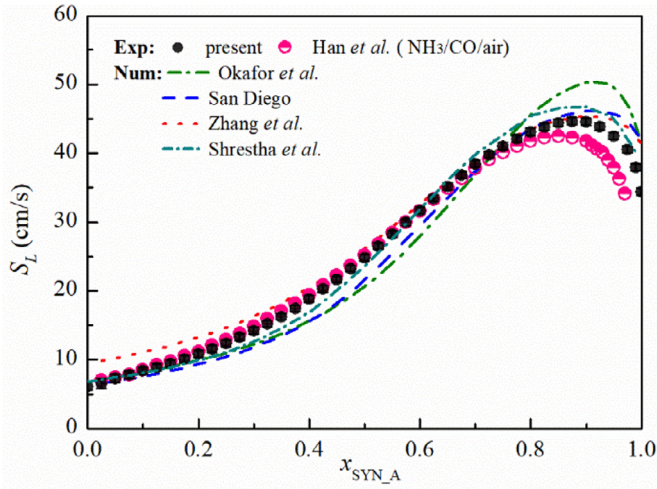


Fig. 1. Laminar burning velocity of stoichiometric NH_3/SYN_A (5 vol% H_2 + 95 vol% CO)/air flames at 1 atm and 298 K with different syngas mixing ratio. NH_3/CO /air results from Han et al. [19] shown for comparison.

with the above 4 mechanisms. There are other NH_3 containing mechanisms, while recent modeling studies on NH_3 flames [4,29] and our previous work [19] demonstrated that their performances are not better than those used here, thus they were not selected. It should be noted that mechanisms by Song et al. [30], Nakamura et al. [31], Otomo et al. [32], and Mei et al. [33] do not include CO oxidation reactions, thus they were only used for cases without CO component. Also, the selected mechanisms except that of Mathieu and Petersen [23], Nakamura et al. [31], Zhang et al. [26], Shrestha et al. [27], and Mei et al. [33] do not include OH_{EX} species, thus they will not be used to calculate the ignition delay times.

3. Results and discussion

3.1. $\text{NH}_3/\text{syngas}/\text{air}$ experiment

3.1.1. $\text{NH}_3/\text{SYN}_A/\text{air}$

Figure 1 shows the measured laminar burning velocities of stoichiometric NH_3/SYN_A (5 vol% H_2 + 95 vol% CO)/air mixtures. Non-monotonic S_L dependence on mixing ratio x_{SYN_A} can be found in Fig. 1, which has also been found in $\text{NH}_3/\text{CO}/\text{air}$ mixtures [19]. Comparison of S_L values of stoichiometric $\text{NH}_3/\text{SYN}_A/\text{air}$ flames in Fig. 1 and $\text{NH}_3/\text{CO}/\text{air}$ flames in [19] reveals that the former and the latter overlap at mixing ratio $x < 0.6$; and only when NH_3 in the mixtures becomes very little does the former become significantly larger than the latter. This phenomena can be understood by the dominance of CO in SYN_A syngas, and the importance of the H radical pool in NH_3 based flames explained in [19]. When the mixing ratio x_{SYN_A} or x_{CO} is little, the H radical pool is controlled by the NH_3 oxidation processes, and whether there is small amount of H_2 substituting CO to provide extra H source makes negligible influence on the size of H pool. While when the mixing ratio x_{SYN_A} or x_{CO} approaches to unity, the oxidation process is governed mostly by CO reactions, which are highly sensitive to the presence of H , and thus small amount of extra H source addition can enhance the burning velocity significantly.

Figure 1 also shows the predicted laminar burning velocities using the four mechanisms. The Okafor et al. mechanism [24] and San Diego mechanism [25] show same tendency as in predicting the $\text{NH}_3/\text{CO}/\text{air}$ flames [19], both of which give under-estimation of S_L at around $x_{\text{SYN}_A} = 0.5$. The Zhang et al. mechanism [26], though agreeing well with experimental results around $x_{\text{SYN}_A} = 0.5$, over predicts by a factor of almost 2 the burning velocity of

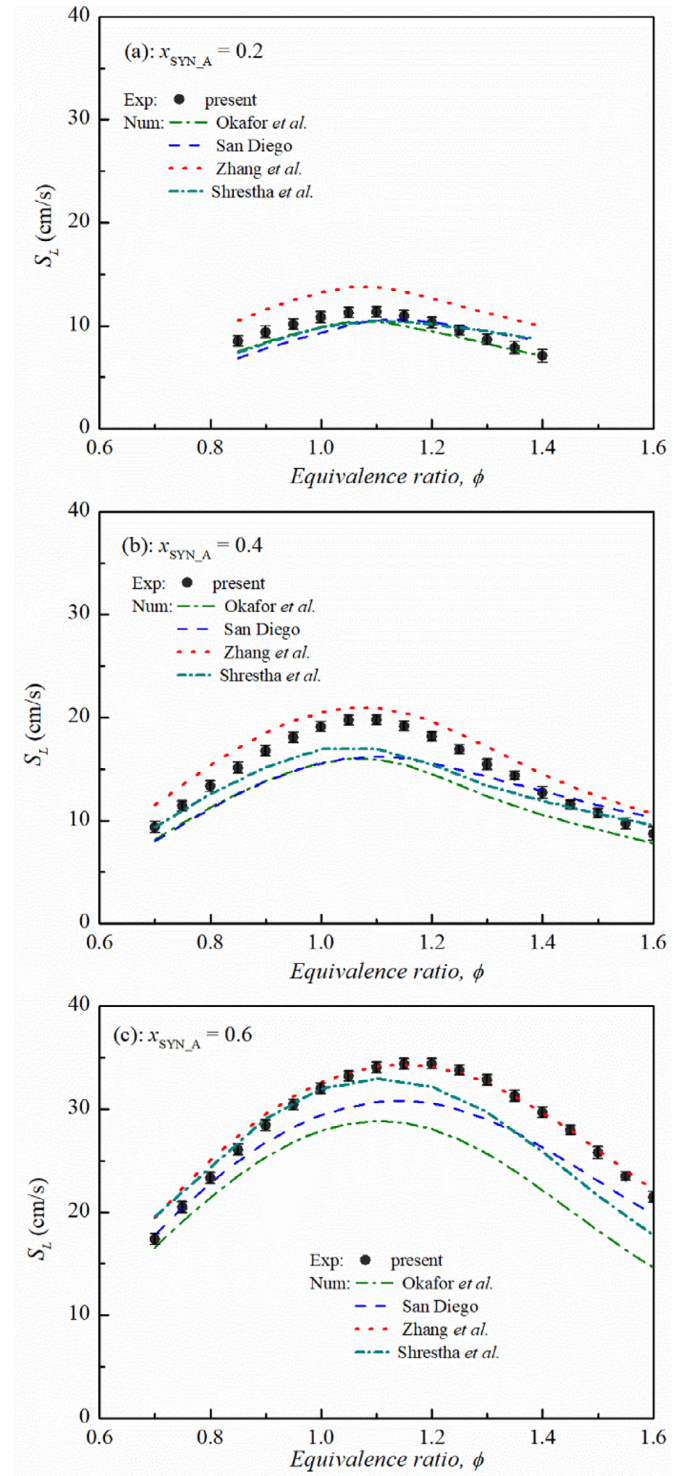


Fig. 2. Laminar burning velocity of NH_3/SYN_A (5 vol% H_2 + 95 vol% CO)/air flames at 1 atm and 298 K with $x_{\text{SYN}_A} = 0.2, 0.4$, and 0.6 .

pure NH_3/air flame. It is also noted that the above three mechanisms over-estimate at $x_{\text{SYN}_A} = 1.0$, indicating that the $\text{H}/\text{C}/\text{O}$ sub-mechanisms used are not very accurate in predicting syngas flames with large percentage of CO . Shrestha et al. [27] reproduces best the experimental results, with small under-estimation and over-estimation found respectively at around $x_{\text{SYN}_A} = 0.4$ and $x_{\text{SYN}_A} = 0.8$.

Figure 2 shows laminar burning velocities of $\text{NH}_3/\text{SYN}_A/\text{air}$ flames, as a function of ϕ , measured and predicted for $x_{\text{SYN}_A} =$

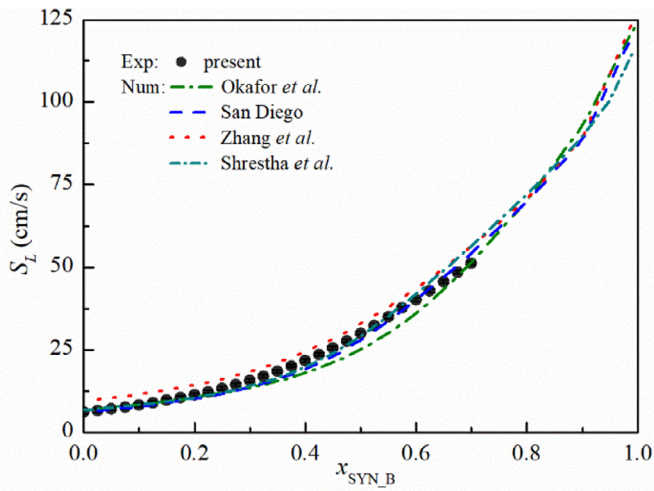


Fig. 3. Laminar burning velocity of stoichiometric NH_3/SYN_B (50 vol% H_2 + 50 vol% CO)/air flames at 1 atm and 298 K with varying syngas mixing ratio.

0.2, 0.4, and 0.6. Same as in the $\text{NH}_3/\text{CO}/\text{air}$ mixtures [19], the value of ϕ with the maximum $\text{NH}_3/\text{SYN}_A/\text{air}$ S_L also increases with the increase in x_{SYN_A} in Fig. 2, and the shift in position of the S_L peak along the equivalence ratio axis is the same, being 1.1 for $x_{\text{SYN}_A} = 0.2$ and 0.4; and 1.2 for $x_{\text{SYN}_A} = 0.6$. This again can be explained by the dominance of CO in SYN_A syngas. Results by mechanisms from Okafor et al. [24], Shrestha et al. [27] and San Diego [25] show better agreement with experimental results at $x_{\text{SYN}_A} = 0.2$ than at $x_{\text{SYN}_A} = 0.4$ and 0.6, where they are lower than the experimental values of S_L for most ϕ values studied. It is also found that the fuel-rich side predictions by San Diego mechanism [25] are always larger than that by Okafor et al. mechanism [24], which may be because of the neglecting of typical fuel-rich species in San Diego mechanism [25], i.e., N_2H_2 , N_2H_3 , and N_2H_4 . The predictions using Zhang et al. mechanism [26] almost overlap with experimental S_L at $x_{\text{SYN}_A} = 0.6$, however, over-estimations are found at lower syngas mixing ratios (0.2 and 0.4).

3.1.2. $\text{NH}_3/\text{SYN}_B/\text{air}$

Figure 3 shows the measured laminar burning velocities of stoichiometric NH_3/SYN_B (50 vol% H_2 + 50 vol% CO)/air mixtures. The dependence of S_L on x_{SYN_B} can be found to be nearly exponential, which is more alike that of $\text{NH}_3/\text{H}_2/\text{air}$ flames [19] than $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{SYN}_A/\text{air}$ flames (Fig. 1). Figure 3 also shows the predicted laminar burning velocities using the four mechanisms. Overall, they perform better than predicting SYN_A in Fig. 1. The Okafor et al. mechanism [24] slightly under-estimates at around $x_{\text{SYN}_B} = 0.5$. Mechanisms by San Diego [25] and Shrestha et al. [27] predict higher S_L than the Okafor et al. [24] in the ranges investigated experimentally, but there is still little underestimations found at around $x_{\text{SYN}_B} = 0.4$. The Zhang et al. mechanism [26] always makes larger predictions than the experimental data.

Figure 4 shows laminar burning velocities of $\text{NH}_3/\text{SYN}_B/\text{air}$ flames, as a function of ϕ , measured and predicted for $x_{\text{SYN}_B} = 0.2, 0.4$, and 0.6. The value of ϕ with maximum $\text{NH}_3/\text{SYN}_B/\text{air}$ S_L increases with the increasing of x_{SYN_B} in Fig. 4, being 1.05 for $x_{\text{SYN}_B} = 0.2$; 1.1 for $x_{\text{SYN}_B} = 0.4$; and 1.15 for $x_{\text{SYN}_B} = 0.6$. Predictions by Okafor et al. mechanism [24] agrees well with experimental results at $x_{\text{SYN}_B} = 0.2$, but are much lower than the experimental results at $x_{\text{SYN}_B} = 0.4$ and 0.6. San Diego mechanism [25] shows good agreement at the fuel-lean and stoichiometric side of $x_{\text{SYN}_B} = 0.6$ flames, but the fuel-rich side predictions are higher than the experimental data; and at lower syngas mixing ra-

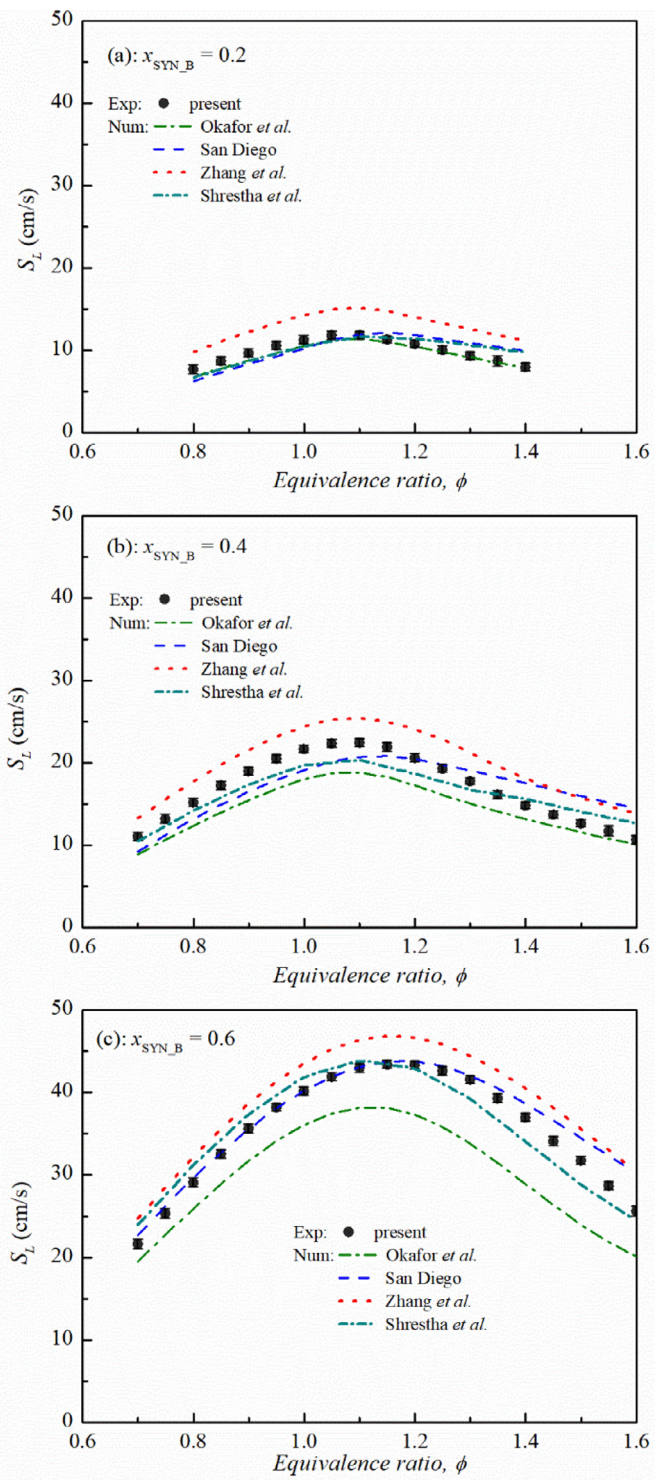


Fig. 4. Laminar burning velocity of NH_3/SYN_B (50 vol% H_2 + 50 vol% CO)/air flames at 1 atm and 298 K with $x_{\text{SYN}_B} = 0.2, 0.4$, and 0.6.

tios, under-estimation at fuel-lean side and over-estimation at fuel-rich side can be found. Different from the $\text{NH}_3/\text{SYN}_A/\text{air}$ results in Fig. 2, the predictions using Zhang et al. mechanism [26] are higher than the experimental S_L at all mixing ratios investigated. Shrestha et al. [27] mechanism under-estimates at around stoichiometric $x_{\text{SYN}_B} = 0.4$ flame, and at the fuel-rich side of $x_{\text{SYN}_B} = 0.6$ flame. In a word, none of the four chosen mechanisms can provide accurate predictions at all cases measured in the present work.

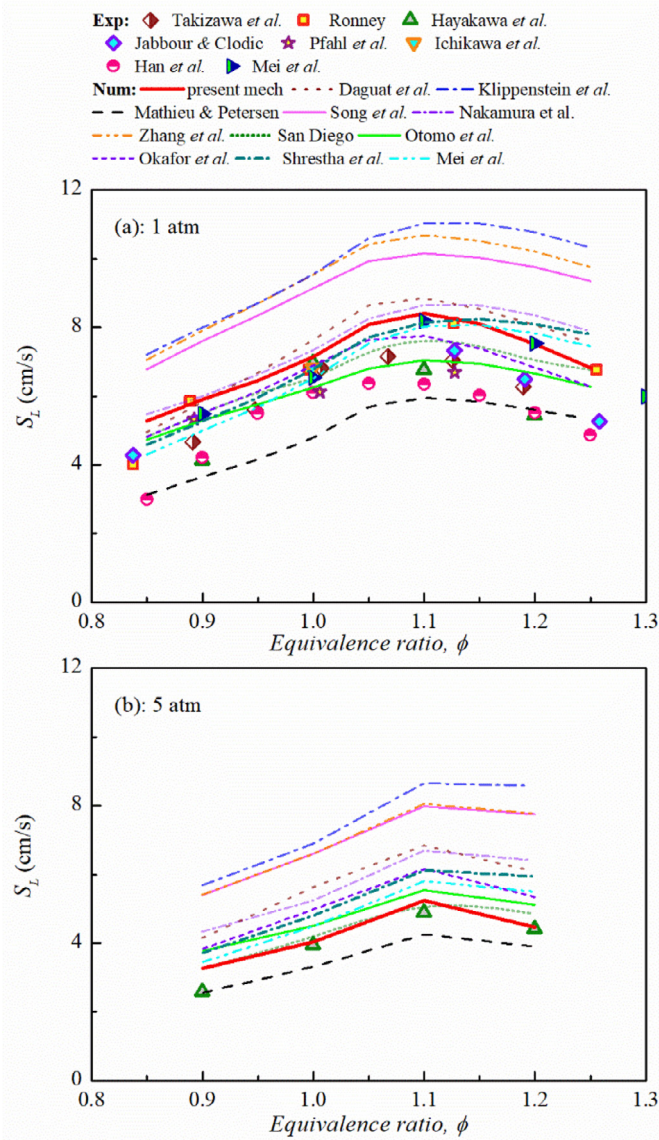


Fig. 5. Laminar burning velocities of NH_3/air flames at 298 K and (a) 1 atm; (b) 5 atm. Literature experimental data are from Han et al. [19], Takizawa et al. [9], Ronney [65], Pfahl et al. [66], Jabbour et al. [67], Hayakawa et al. [68] and Ichikawa et al. [69].

3.2. $\text{NH}_3/\text{syngas}/\text{air}$ mechanism

3.2.1. Mechanism development

According to Section 3.1 and [19], none of the existing mechanism in literature can satisfy the predictions of adiabatic burning velocities of $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{syngas}/\text{air}$ flames at all different conditions. To solve this problem, a mechanism targeting at these systems has been developed, which is described below.

Syngas sub-mechanism: In the present mechanism, the syngas mechanism by Varga et al. [14], namely ELTE-syngas mechanism, is used as the $\text{H}/\text{C}/\text{O}$ subset. Varga et al. [14] optimized this mechanism by a mathematical technique and the validation was made against comprehensive sets of experimental data. Review work by Curran [36] commented on the optimization works done by Varga and coworkers as ‘recent successes’, and the burning velocity research by Alekseev and Konnov [37] has shown the calculating results by ELTE-syngas mechanism agree better with experimental data than other peer mechanisms.

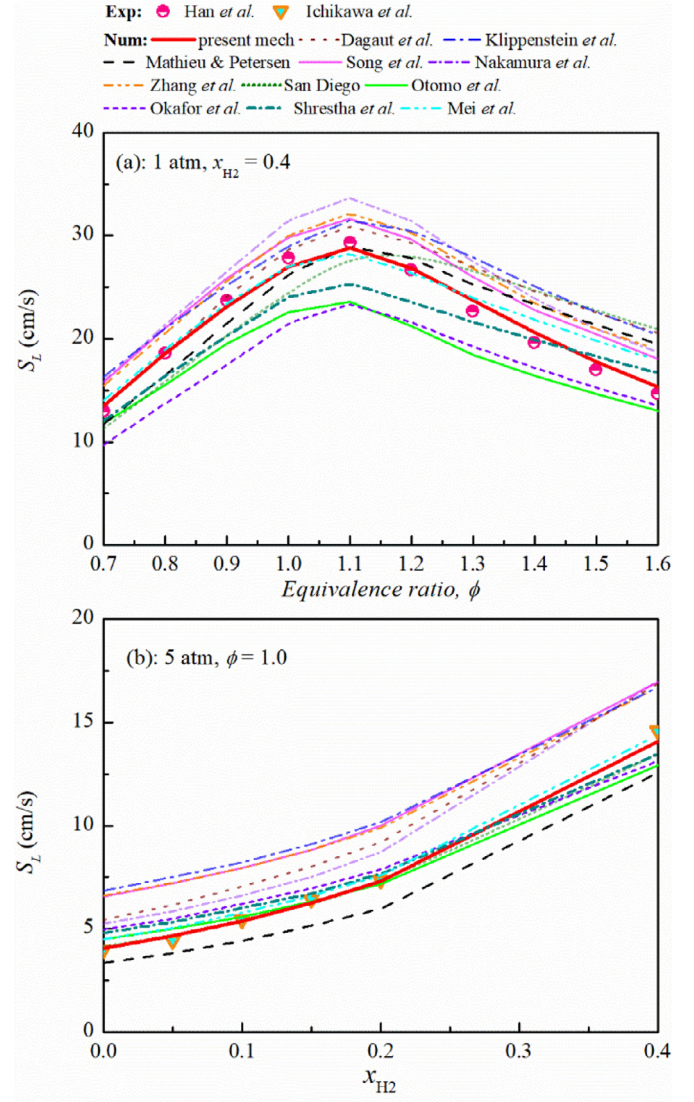


Fig. 6. Laminar burning velocities of $\text{NH}_3/\text{H}_2/\text{air}$ flames at 298 K and (a) 1 atm with $x_{\text{H}_2} = 0.4$; (b) 5 atm at stoichiometry. Literature experimental data are from Han et al. [19] and Ichikawa et al. [69].

NH_3 sub-mechanism: The NH_3 sub-mechanism in the present study is organized in the following manner.

- (1) N element containing reactions were collected from existing NH_3 containing mechanisms and literature researches on elementary reactions.
- (2) Our previous publication [19] and the study from Okafor et al. [24] suggest that the interaction between C containing species and N containing species are not important for the NH_3 blending flames, thus reactions and species that have C and N element at the same time were deleted from the collection in (1).
- (3) In the residual of reaction collection, there are some reactions, where for each of them, there are different selections of rate constant parameters, pressure dependence coefficients, or collision factors from different sources. These reactions were carefully checked and for each reaction, the individual selection was tested independently its impact on laminar burning velocity and ignition delay time. All the S_L experimental data shown in [19] and Section 3.1 as well as the ignition delay time data by Mathieu and Petersen

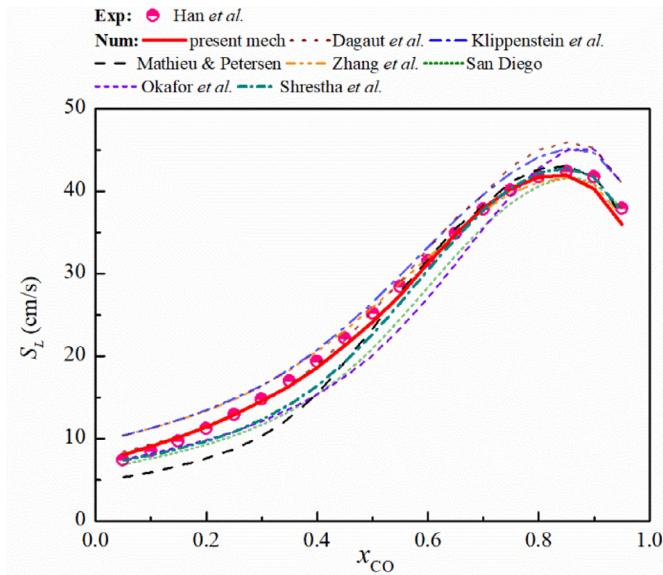


Fig. 7. Laminar burning velocities of stoichiometric $\text{NH}_3/\text{CO}/\text{air}$ flames at 1 atm and 298 K. Literature experimental data are from Han et al. [19].

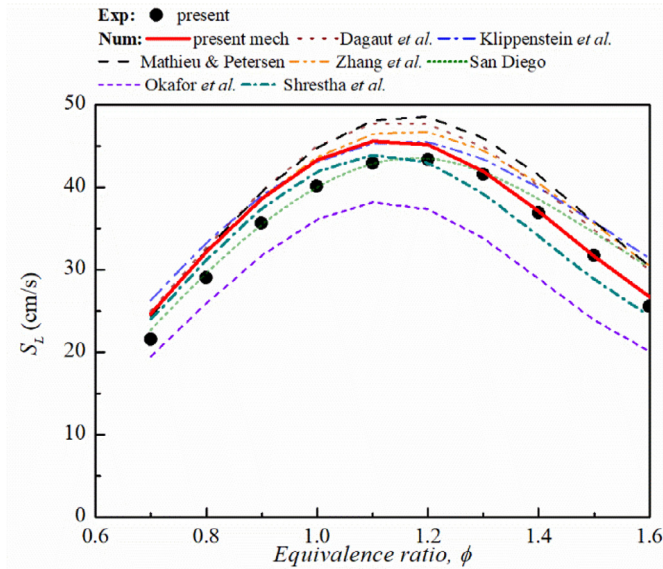


Fig. 8. Laminar burning velocities of NH_3/SYN_B (50 vol% H_2 + 50 vol% CO)/air flames at 1 atm and 298 K with $x_{\text{SYN}_B} = 0.6$, present mechanism comparing with literature mechanisms and present experimental data.

[23] and Mathieu et al. [15] were calculated at each turn. After going through the reactions in every possible combination of selections, a scheme agreeing well with all tested burning velocity and ignition delay time experimental data was obtained. It should be noted that the present optimizing processes didn't make any tuning to the reaction parameters reported by their original sources.

- (4) The size of present sub-mechanism was kept to minimal after careful reaction sensitivity analyses and path analyses of the obtained scheme, deleting reactions with negligible influence. In this way, the NH_3 sub-mechanism with 20 species and 130 reactions was made and the reactions included are shown in Table 3. It should be noted that the HONO , HNOH , HON , HNO_2 , HONO_2 , and NO_3 reactions are all from Glarborg et al. [38], and this part of reactions is found not influence the predictions of the burning veloc-

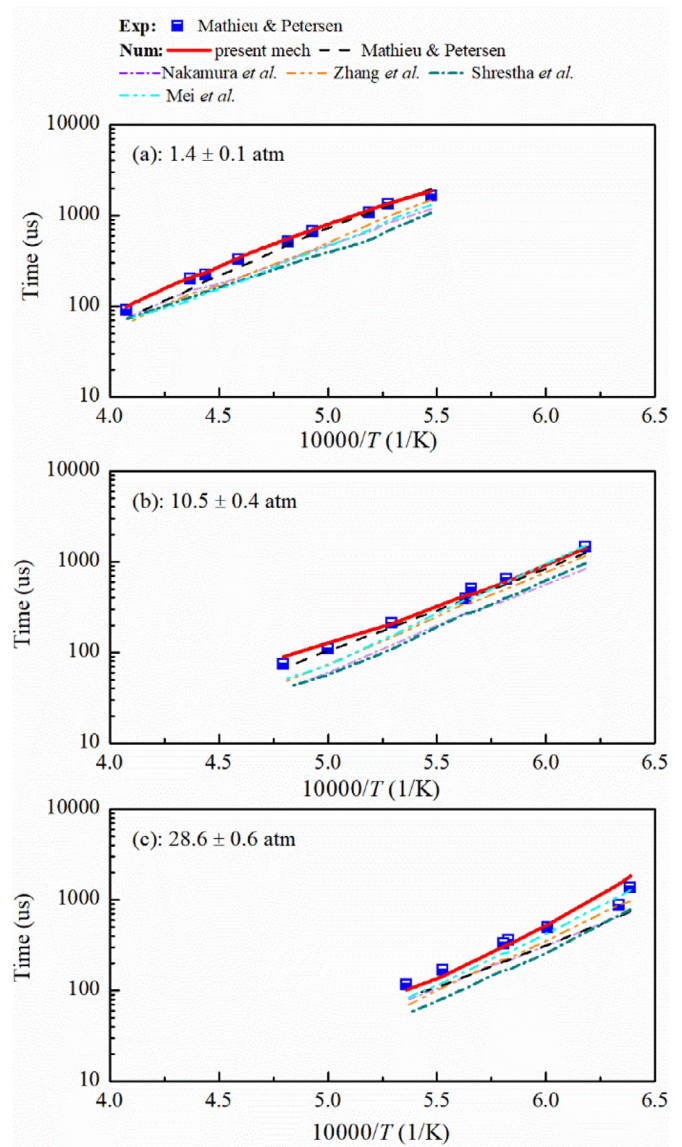


Fig. 9. Ignition delay times of stoichiometric $\text{NH}_3/\text{O}_2/\text{Ar}$ flames (0.01143 $\text{NH}_3/0.00857 \text{ O}_2/0.98 \text{ Ar}$) at around (a) 1.4 atm; (b) 10.5 atm; and (c) 28.6 atm. Literature experimental data are from Mathieu and Petersen [23].

ity and ignition delay time of NH_3 , NH_3/H_2 , NH_3/CO , and $\text{NH}_3/\text{syngas}$ mixtures.

Thermodynamic and transport properties: In the present mechanism, thermodynamic data of all species (including species in the syngas subset) are from Glarborg et al. [38], which provides the most recent collection of thermal dynamic properties. The transport property data are also obtained from Glarborg et al. [38].

3.2.2. Mechanism validation

To show the validity of the present mechanism, several typical conditions from the present and literature experiments are selected in the following.

Figure 5 shows the performance of the present mechanism on predicting laminar burning velocities of NH_3/air flames at 1 atm and 5 atm. The literature experimental data at 1 atm are from Han et al. [19], Takizawa et al. [9], Ronney [65], Pfahl et al. [66], Jabbour et al. [67], Hayakawa et al. [68] and Ichikawa et al. [69]; whereas literature experimental data at 5 atm are from Hayakawa et al. [68]. Among the literature mechanisms, the San

Table 3

Reactions in the present NH₃ sub-mechanism. Important reactions that have different rate parameters compared from different sources are label * in the last column.

No.	Reaction	Reaction rate (cm, mol, s, cal)			Ref.	Notes
		A	n	E		
R1	NH ₃ + O = NH ₂ + OH	1.63×10^7	1.85	6457	[39]	*
R2	NH ₃ + H = NH ₂ + H ₂	1.26×10^{14}	0	21,500	[40]	*
R3	NH ₃ + OH = NH ₂ + H ₂ O	5.0×10^7	1.6	954	[41]	*
R4	NH ₃ + HO ₂ = NH ₂ + H ₂ O ₂	3.0×10^{11}	0	22,000	[42]	*
R5	NH ₃ + NO = NH ₂ + HNO	3.26×10^7	1.73	56,600	[43]	*
R6	NH ₃ + NH + M = N ₂ H ₄ + M	1.09×10^{15}	0	0	[44]	*
R7	NH ₂ + O = NH + OH DUP	7.0×10^{12} 8.6×10^{-1}	0 4.01	0 1673	[38]	*
R8	HNO + H = NH ₂ + O	3.5×10^{15}	-0.3	29,200	[41]	*
R9	NH + H ₂ = NH ₂ + H	2.1×10^{13}	0	15,417	[45]	*
R10	NH ₂ + OH = NH + H ₂ O	9.6×10^6	1.97	669	[46]	*
R11	NH ₂ + O ₂ = H ₂ NO + O	2.6×10^{11}	0.4872	29,050	[35]	*
R12	NH ₂ + O ₂ = HNO + OH	2.9×10^{-2}	3.764	18,185	[35]	*
R13	NH ₂ + H ₂ NO = HNO + NH ₃	3.0×10^{12}	0	1000	[47]	*
R14	NH ₂ + NO ₂ = N ₂ O + H ₂ O	1.6×10^{16}	-1.44	268	[48]	*
R15	NH ₂ + NO ₂ = H ₂ NO + NO	6.5×10^{16}	-1.44	268	[48]	*
R16	NH ₂ + NO = N ₂ + H ₂ O	2.6×10^{19}	-2.369	870	[49]	*
R17	NH ₂ + NO = NNH + OH	4.3×10^{10}	0.294	-866	[49]	*
R18	H ₂ NO + H = NH ₂ + OH	5.0×10^{13}	0	0	[47]	*
R19	NH ₂ + HO ₂ = H ₂ NO + OH	2.5×10^{17}	-1.28	1166	[50]	*
R20	NH ₂ + HO ₂ = NH ₃ + O ₂	4.5×10^{13}	0	0	[51]	*
R21	NH ₂ + NH = NH ₃ + N	9.6×10^3	2.46	107	[52]	*
R22	NH ₂ + NH = N ₂ H ₃	7.0×10^{13}	0	0	[53]	*
R23	NH ₂ + NH = N ₂ H ₂ + H	4.3×10^{14}	-0.272	-77	[52]	*
R24	NH ₂ + NH ₂ = NH ₃ + NH	5.6×10^0	3.53	552	[52]	*
R25	NH ₂ + NH ₂ (+M) = N ₂ H ₄ (+M) LOW TROE	5.6×10^{14} 1.6×10^{34} 3.1×10^{-1}	-0.414 -5.49 1×10^{-30}	66 1987 1×10^{30}	[52]	*
R26	NH ₂ + NH ₂ = N ₂ H ₃ + H	1.2×10^{12}	-0.03	10,084	[54]	*
R27	NH ₂ + NH ₂ = N ₂ H ₂ + H ₂	1.7×10^8	1.62	11,783	[52]	*
R28	NH ₂ + NH ₂ = H ₂ NN + H ₂	7.2×10^4	1.88	8802	[52]	*
R29	NH ₂ + N = N ₂ + H + H	7.2×10^{13}	0	0	[47]	*
R30	NH + H = N + H ₂	3.01×10^{13}	0	0	[55]	*
R31	NH + N = N ₂ + H	9.0×10^{11}	0.5	0	[56]	*
R32	NH + OH = N + H ₂ O	2.0×10^9	1.2	0	[41]	*
R33	NH + OH = HNO + H	3.2×10^{14}	-0.376	-46	[52]	*
R34	NH + O ₂ = HNO + O	2.4×10^{13}	0	13,850	[38]	*
R35	NH + O ₂ = NO + OH	2.014×10^{16}	-1.38	5672	[57]	*
R36	NH + NO = N ₂ O + H	1.8×10^{14}	-0.351	-244	[35]	*
R37	NH + NO = N ₂ + OH	2.7×10^{12}	-0.0721	-512	[35]	*
R38	NH + NH = NH ₂ + N	5.7×10^{-1}	3.88	342	[52]	*
R39	NH + NH = N ₂ H ₂	6.26×10^{13}	-0.036	-161	[52]	*
R40	N + O + M = NO + M H ₂ O /16.25/	7.6×10^{14}	-0.1	-1770	[26]	*
R41	N + NO = N ₂ + O	2.1×10^{13}	0	0	[39]	*
R42	N + O ₂ = NO + O	5.9×10^9	1.0	6199	[39]	*
R43	N ₂ + M = N + N + M H ₂ O /16.25/	1.89×10^{18}	-0.85	224,950	[26]	*
R44	NNH = N ₂ + H	1.0×10^9	0	0	[35]	*
R45	NNH + O = N ₂ + OH	1.2×10^{13}	0.145	-217	[35]	*
R46	NNH + O = N ₂ O + H	1.0×10^{14}	0	0	[47]	*
R47	NNH + O = NH + NO	3.3×10^{14}	-0.23	-1013	[58]	*
R48	NNH + OH = N ₂ + H ₂ O	5.0×10^{13}	0	0	[47]	*
R49	NNH + H = N ₂ + H ₂	1.0×10^{12}	0	0	[58]	*
R50	NNH + O ₂ = N ₂ + HO ₂	5.6×10^{14}	-0.385	-13	[35]	*
R51	NNH + O ₂ = N ₂ + H + O ₂	5.0×10^{13}	0	0	[47]	*
R52	NNH + NO = N ₂ + HNO	5.0×10^{13}	0	0	[47]	*
R53	N ₂ H ₂ + O = NNH + OH	3.3×10^8	1.5	497	[54]	*
R54	N ₂ H ₂ + H = NNH + H ₂	1.1×10^{14}	0	3128	[38]	*
R55	N ₂ H ₂ + OH = NNH + H ₂ O	5.9×10^1	3.4	-1360	[59]	*
R56	N ₂ H ₂ + NH = NNH + NH ₂	2.4×10^6	2.0	-1192	[54]	*
R57	N ₂ H ₂ + NO = N ₂ O + NH ₂	4.0×10^{12}	0	11,922	[54]	*
R58	N ₂ H ₂ + NH ₂ = NNH + NH ₃	8.8×10^{-2}	4.05	-1610	[59]	*
R59	H ₂ NN + H = NNH + H ₂	4.8×10^8	1.5	-894	[54]	*
R60	H ₂ NN + H = N ₂ H ₂ + H	7.0×10^{13}	0	0	[54]	*
R61	H ₂ NN + O = NNH + OH	3.3×10^8	1.5	-894	[54]	*
R62	H ₂ NN + O = NH ₂ + NO	7.0×10^{13}	0	0	[54]	*
R63	H ₂ NN + OH = NNH + H ₂ O	2.4×10^6	2.0	-1192	[54]	*
R64	H ₂ NN + HO ₂ = NNH + H ₂ O ₂	2.9×10^4	2.69	-1600	[54]	*
R65	H ₂ NN + O ₂ = NH ₂ + NO ₂	1.5×10^{12}	0	5961	[54]	*
R66	H ₂ NN + NH ₂ = NNH + NH ₃	1.8×10^6	1.94	-1152	[54]	*

(continued on next page)

Table 3 (continued)

No.	Reaction	Reaction rate (cm, mol, s, cal)			Ref.	Notes
		A	n	E		
R67	$\text{N}_2\text{H}_3 + \text{O} = \text{N}_2\text{H}_2 + \text{OH}$	1.7×10^8	1.5	−646	[54]	
R68	$\text{N}_2\text{H}_3 + \text{O} = \text{NH}_2 + \text{HNO}$	3.0×10^{13}	0	0	[54]	
R69	$\text{N}_2\text{H}_3 + \text{H} = \text{N}_2\text{H}_2 + \text{H}_2$	2.4×10^8	1.5	−10	[54]	*
R70	$\text{N}_2\text{H}_3 + \text{OH} = \text{N}_2\text{H}_2 + \text{H}_2\text{O}$	1.2×10^6	2.0	−1192	[54]	
R71	$\text{N}_2\text{H}_3 + \text{NH}_2 = \text{N}_2\text{H}_2 + \text{NH}_3$	9.2×10^5	1.94	−1152	[54]	
R72	$\text{N}_2\text{H}_4 + \text{H} = \text{N}_2\text{H}_3 + \text{H}_2$	4.5×10^7	1.8	2613	[60]	*
R73	$\text{N}_2\text{H}_4 + \text{OH} = \text{N}_2\text{H}_3 + \text{H}_2\text{O}$	4.0×10^{13}	0	0	[61]	*
R74	$\text{HNO} + \text{O} = \text{NO} + \text{OH}$	3.61×10^{13}	0	0	[62]	*
R75	$\text{HNO} + \text{H} = \text{NO} + \text{H}_2$	4.47×10^{11}	0.72	650	[63]	*
R76	$\text{HNO} + \text{OH} = \text{NO} + \text{H}_2\text{O}$	3.6×10^{13}	0	0	[26]	*
R77	$\text{HNO} + \text{N} = \text{NO} + \text{NH}$	1.0×10^{13}	0	1990	[23]	
R78	$\text{NO} + \text{O} (+\text{M}) = \text{NO}_2 (+\text{M})$	1.3×10^{15}	−0.75	0	[26]	*
	LOW	9.44×10^{24}	−2.87	1551		
	TROE	0.962 10.0 7962				
	AR /0.6/ NO ₂ /6.2/ NO /1.8/ O ₂ /0.8/ N ₂ O /4.4/ CO ₂ /6.0/ H ₂ O /10.0/					
R79	$\text{NO} + \text{H} = \text{N} + \text{OH}$	2.17×10^{14}	0	49,500	[64]	*
R80	$\text{NO} + \text{H} = \text{NH} + \text{O}$	9.9×10^{14}	−0.1	69,900	[41]	*
R81	$\text{NO} + \text{H} + \text{M} = \text{HNO} + \text{M}$	3.0×10^{20}	−1.75	0	[23]	*
	H ₂ O /4.1/ H ₂ /1.25/					
R82	$\text{NO} + \text{HO}_2 = \text{NO}_2 + \text{OH}$	2.1×10^{12}	0	−497	[39]	*
R83	$\text{N}_2\text{O} + \text{H} = \text{N}_2 + \text{OH}$	6.4×10^7	1.835	13,492	[35]	*
R84	$\text{N}_2\text{O} (+\text{M}) = \text{N}_2 + \text{O} (+\text{M})$	1.69×10^{11}	0	57,653	[26]	*
	LOW	7.2×10^{14}	0	57,410		
	O ₂ /1.4/ N ₂ /1.7/ H ₂ O /12.0/ NO /3.0/ N ₂ O /3.5/					
R85	$\text{NO}_2 + \text{H} = \text{NO} + \text{OH}$	1.507×10^{14}	0	362	[26]	*
R86	$\text{NO}_2 + \text{O} = \text{NO} + \text{O}_2$	5.86×10^{12}	0	−238	[26]	*

R87–R130 are HONO, HNOH, HON, HNO₂, HONO₂, and NO₃ reactions from Glarborg et al. [38]

Diego [25] is found to make the best predictions. Mechanisms from Zhang et al. [26], Klippenstein et al. [35], and Song et al. [30] predict 1.5–2 times the measured burning velocities. Other literature mechanisms except Mathieu and Petersen [23] and San Diego [25], predict lower burning velocity compared to the aforementioned three mechanisms at 1 atm while over-estimating the experimental results at 5 atm. The Mathieu and Petersen mechanism [23] under-estimates stoichiometric burning velocities at both pressures. It is noted that the present mechanism reproduces well the literature experimental data at both atmosphere and elevated pressures.

Figure 6 shows the good performance of the present mechanism on predicting laminar burning velocities of NH₃/H₂/air flames at 1 atm and 5 atm. The literature experimental data at 1 atm are from Han et al. [19], and that at 5 atm are from Ichikawa et al. [69]. It should be noted that the data of Ichikawa et al. [69] was measured for stoichiometric flames, thus the x-axis in Fig. 6 (b) is mixing ratio x_{H_2} rather than equivalence ratio in Fig. 6 (a). None of the tested literature mechanism can predict accurately. At 1 atm, the literature mechanisms except that by Okafor et al. [24], Shrestha et al. [27], and Otomo et al. [32] over-estimate the burning velocities at the fuel-rich side; while these three mechanisms under-estimate for at least 5 cm/s at around stoichiometry. At 5 atm, the literature mechanisms except Mei et al. [33] predict either higher or lower values than the experimental data at $x_{\text{H}_2} = 0.4$.

Figure 7 shows the good performance of the present mechanism on predicting laminar burning velocities of stoichiometric NH₃/CO/air flames at 1 atm. The literature experimental data are from Han et al. [19]. None of the tested literature mechanisms can predict accurately. Mechanisms from Okafor et al. [24], San Diego [25], Mathieu and Petersen [23], and Shrestha et al. [27] under-estimate the S_L at around $x_{\text{CO}} = 0.4$ for at least 4 cm/s; Zhang et al. mechanism [26] and Klippenstein et al. mechanism [35] over-estimate the S_L at around $x_{\text{CO}} = 0$ for around 3 cm/s. The Dagaut et al. mechanism [34] over-estimates the burning velocities at $x_{\text{CO}} > 0.7$.

Figure 8 shows the performance of the present mechanism on predicting laminar burning velocities of stoichiometric NH₃/SYN_B/air flames at $x_{\text{SYN}_B} = 0.6$ and 1 atm. The present experimental data are used for comparison. Mechanism performance of Okafor et al. [24], San Diego [25], Zhang et al. [26], and Shrestha et al. [27] has been discussed in Fig. 4. The other literature mechanisms over-estimate the S_L for all the equivalence ratio range. It is noted that the present mechanism agrees well with the experimental data at the fuel rich side, with the stoichiometric and fuel-lean side predictions being slightly higher.

Figure 9 shows the performance of the present mechanism on predicting ignition delay times for NH₃/O₂/Ar mixtures at around 1.4 atm, 10.5 atm, and 28.6 atm. The literature experimental data are from Mathieu and Petersen [23]. Simulation results by all the literature mechanisms except Mathieu and Petersen [23] show lower value than most experimental data. Results by the Mathieu and Petersen mechanism [23] agree well with the experimental data at 1.4 atm and 10.5 atm, but showing underestimation at 28.6 atm. As for the present mechanism, good agreement can be found for all pressures investigated.

Figure 10 shows the performance of the present mechanism on predicting ignition delay times of NH₃/CO/H₂/O₂/Ar mixtures at around 1.5 atm, 12.0 atm, and 30.0 atm. The syngas mixing ratio is $x_{\text{SYN}_B} = 0.98$ for the shown literature experimental results from Mathieu et al. [15], which was to investigate the influence of impurities on the combustion of syngas. The present mechanism and mechanisms from Zhang et al. [26] and Mathieu and Petersen [23] all reproduce well the experimental data. Shrestha et al. [27] mechanism under-estimates for 40 % the ignition delay time at the low temperature range of 30 atm.

Figure 11 shows the performance of the present mechanism on predicting NO and N₂O species profile of burner stabilized NH₃/NO/Ar flame, literature experimental data are from Vandooren et al. [70]. Simulation results were calculated using the temperature profile provided by Vandooren et al. [70]. The present mechanism was not optimized for the NO_x emission characteristics (see

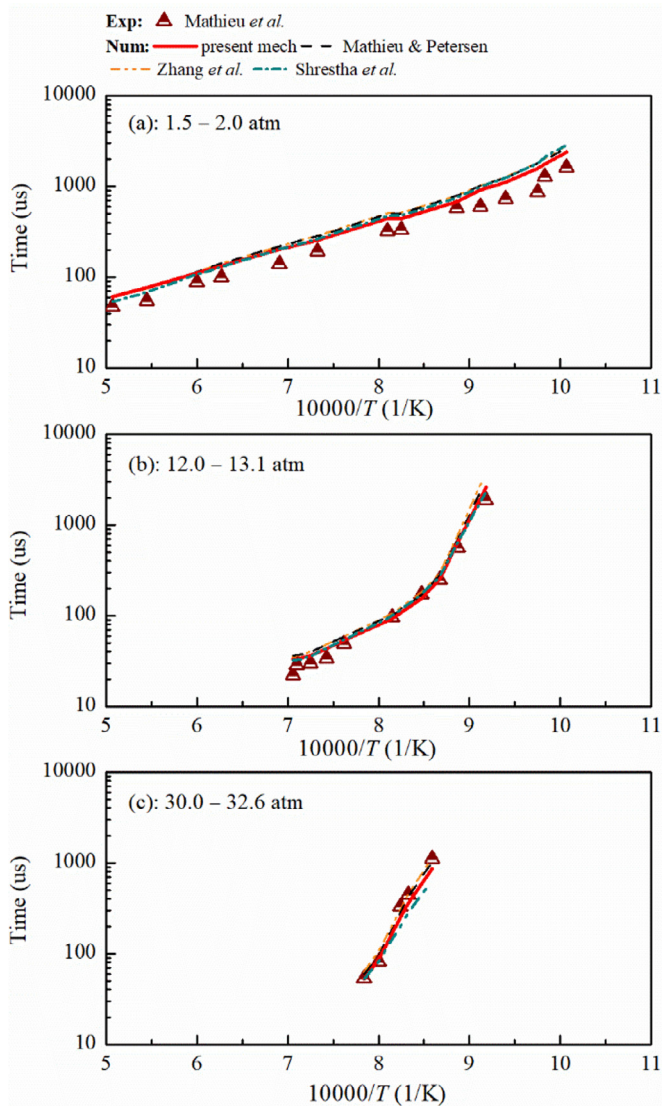


Fig. 10. Ignition delay times of $\phi = 0.515$, $x_{\text{SYN}_B} = 0.98$ $\text{NH}_3/\text{CO}/\text{H}_2/\text{O}_2/\text{Ar}$ flames (0.0002 NH_3 /0.005 H_2 /0.005 CO /0.01 O_2 /0.9798 Ar) at around (a) 1.5 atm; (b) 12.0 atm; and (c) 30.0 atm. Literature experimental data are from Mathieu et al. [15].

NH_3 sub-mechanism), however, the calculation results using present mechanism match the trend of measured NO and N_2O profile. As for the comparison with literature mechanisms, predictions by Shrestha et al. [27], Mei et al. [33], Dagaut et al. [34], Okafor et al. [24], and San Diego [25] are more close to the NO experimental data at around distance = 0.75 cm than the present mechanism, while the present mechanism reproduces better the N_2O maximum mole fraction. To find out more, validations of present mechanism against ignition delay time and JSR species profile of NO, N_2O , and NO_2 systems are available in Supplementary material.

From the above discussion, results calculated by the present mechanism is suitable to simulate NH_3 , NH_3/H_2 , NH_3/CO , and $\text{NH}_3/\text{syngas}$ systems.

3.2.3. Kinetic analyses

Since the present mechanism reproduces well the experimental results, it was used to analyze the oxidation processes of $\text{NH}_3/\text{syngas}/\text{air}$ flames.

Figure 12 shows the NH_3 oxidation paths of the stoichiometric $\text{NH}_3/\text{SYN}_A/\text{air}$ and $\text{NH}_3/\text{SYN}_B/\text{air}$ flames for $x_{\text{SYN}} = 0.4$ computed using the present mechanism. For a certain reaction path, thickness

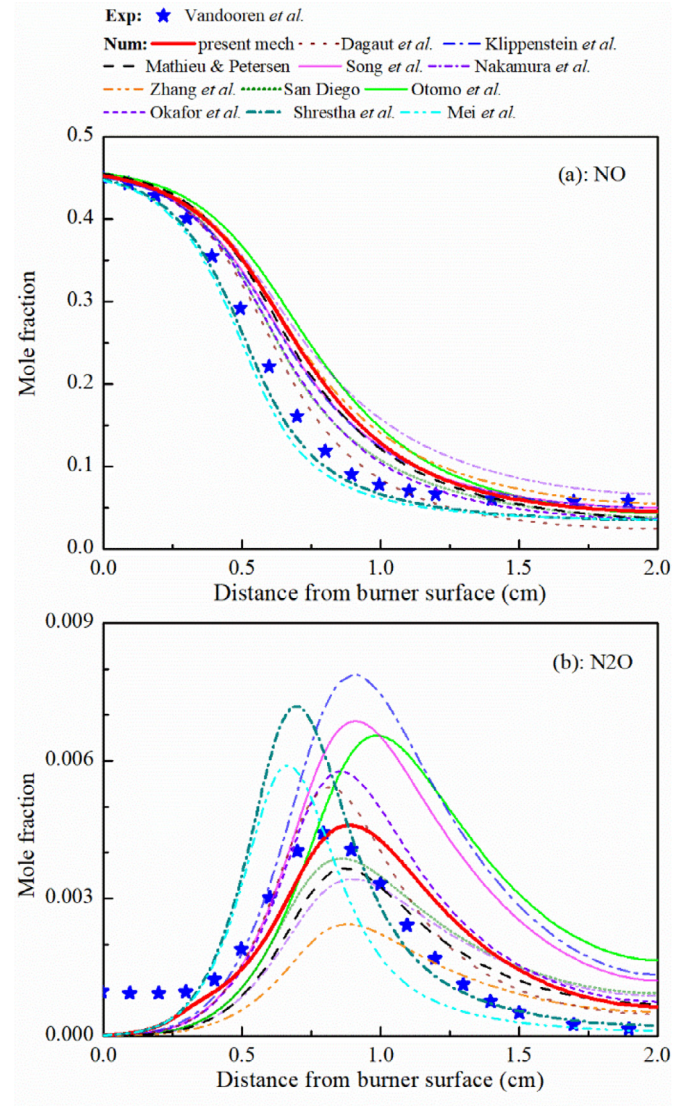


Fig. 11. Spatial distribution of NO and N_2O species in burner stabilized $\text{NH}_3/\text{NO}/\text{Ar}$ flame (0.461 NH_3 /0.472 NO /0.067 Ar , mole basis). Literature experimental data are from Vandooren et al. [70].

of the arrows in Fig. 12 represents the integrated rate of species production (ROP) normalized by that of $\text{H} + \text{O}_2 = \text{OH} + \text{O}$. This way of making reaction path plots is same with the previous publication [19]. All reactions with normalized integrated ROP larger than 5 % are represented by solid lines, otherwise by dashed lines.

From Fig 12, it can be found that the normalized NH_3 reaction paths are nearly the same for $\text{NH}_3/\text{SYN}_A/\text{air}$ and $\text{NH}_3/\text{SYN}_B/\text{air}$ flames, where the oxidation processes begin with NH_3 reacting with OH radical and H atom, and the reaction with OH (R3) is the predominant route. Following the solid lines, the afterward NH_2 oxidation can be categorized into the three routes: (1) $\text{NH}_2\text{--NH--N--N}_2$; (2) $\text{NH}_2\text{--HNO--NO--N}_2$; (3) $\text{NH}_2/\text{NH--(N}_2\text{H}_3\text{--)N}_2\text{H}_2\text{--NNH--N}_2$, shown respectively at the middle, the left, and the right of each subplot, which are similar to the reaction paths calculated by the Okafor et al. mechanism [24] for $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$ flames [19]. There are species that attach to these routes by dashed lines, where the H_2NO , NO_2 , and N_2O can be considered as the additional components of route (2); while N_2H_4 and H_2NN the additional components of route (3). An extensive reaction path analyses (not shown here) found that when it moves from the fuel-lean to stoichiometric and to fuel-rich side

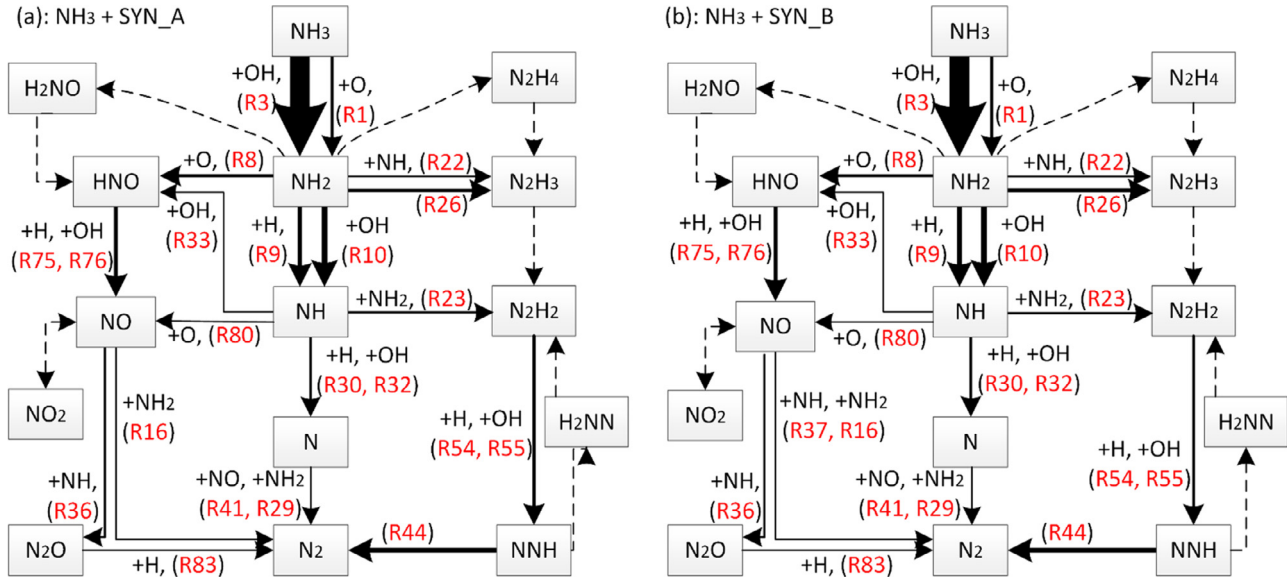


Fig. 12. Reaction paths of the N containing species for stoichiometric $\text{NH}_3/\text{SYN}_\text{A}/\text{air}$ and $\text{NH}_3/\text{SYN}_\text{B}/\text{air}$ flames for $x_{\text{SYN}} = 0.4$, computed using the present mechanism.

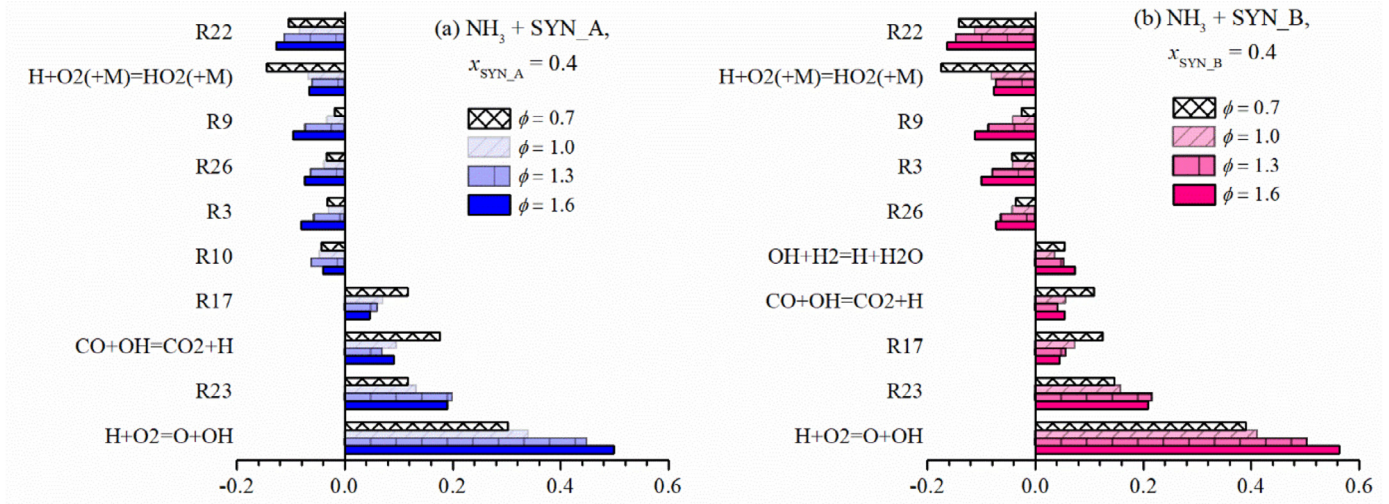


Fig. 13. Normalized sensitivity coefficients of the burning velocity of the 10 most rate limiting reactions in $\text{NH}_3/\text{SYN}_\text{A}/\text{air}$ and $\text{NH}_3/\text{SYN}_\text{B}/\text{air}$ flames for $x_{\text{SYN}} = 0.4$ and $\phi = 0.7, 1.0, 1.3$ and 1.6 , computed using the present mechanism.

of $\text{NH}_3/\text{syngas}/\text{air}$ flames, the integrated rate of species production for reactions in route (2) have decreasing tendency, while that in routes (1) and (3) increase with the latter being more profound than the former. Thus route (2) can be understood as the typical fuel-lean oxidation route and route (3) typical fuel-rich route.

Figure 13 shows the normalized sensitivity coefficients of the burning velocity of the 10 most rate-limiting reactions in $\text{NH}_3/\text{SYN}_\text{A}/\text{air}$ and $\text{NH}_3/\text{SYN}_\text{B}/\text{air}$ flames for $x_{\text{SYN}} = 0.4$ and $\phi = 0.7, 1.0, 1.3$ and 1.6 , computed using the present mechanism. It can be found that the reaction $\text{H} + \text{O}_2 = \text{O} + \text{OH}$ has always the largest positive sensitivity. The most rate-limiting reactions in the NH_3 sub-mechanism are (R3), (R9), (R17), (R22), (R23), and (R26).

Among these reactions, (R3) is the predominant NH_3 consuming reaction in Fig. 12, and (R9) can be found one of the main paths of NH_2 forming NH in route (1). Also, reactions (R22), (R23), and (R26) are important paths attaching NH_2 and NH with N_2H_3 and N_2H_2 in route (3). Reaction (R17) is a chain-branching reaction, and there is a chain-terminating reaction (R16) sharing same reactants with (R17). It has been found that the predicted burning

velocities and NO_x emission are very sensitive to the branching ratio of these two reactions $\alpha = k_{\text{R17}}/(k_{\text{R17}} + k_{\text{R16}})$ [4,38], thus the high S_L -sensitivity value of (R17) can be understood.

By similar kinetic analyses using different mechanisms, it can be understood why the present mechanism introduces no new kinetic information on the already well-known NH_3 oxidation chemistry while reproducing better the experimental results than literature mechanisms. Example of the A-factor reaction sensitivity for $x_{\text{SYN}_\text{B}} = 0.6$ and $\phi = 1$ flame is shown in Fig. 14, where the 10 most rate-limiting reactions are not the same for all mechanisms; and for a certain reaction, different mechanisms report different sensitivity coefficients. The different performances in reaction sensitivities are in line with the discrepancies among simulated S_L by different mechanisms in Fig. 8, which is attributed mostly to the differences in reactions and rate parameters (also selection of different thermodynamic or transport property data). Figure 15 shows an example of how big differences are there in the branching ratio $\alpha = k_{\text{R17}}/(k_{\text{R17}} + k_{\text{R16}})$ in different mechanisms, where around 1800 K, the α chosen by Shrestha et al. [27], and Mei et al. [33] is

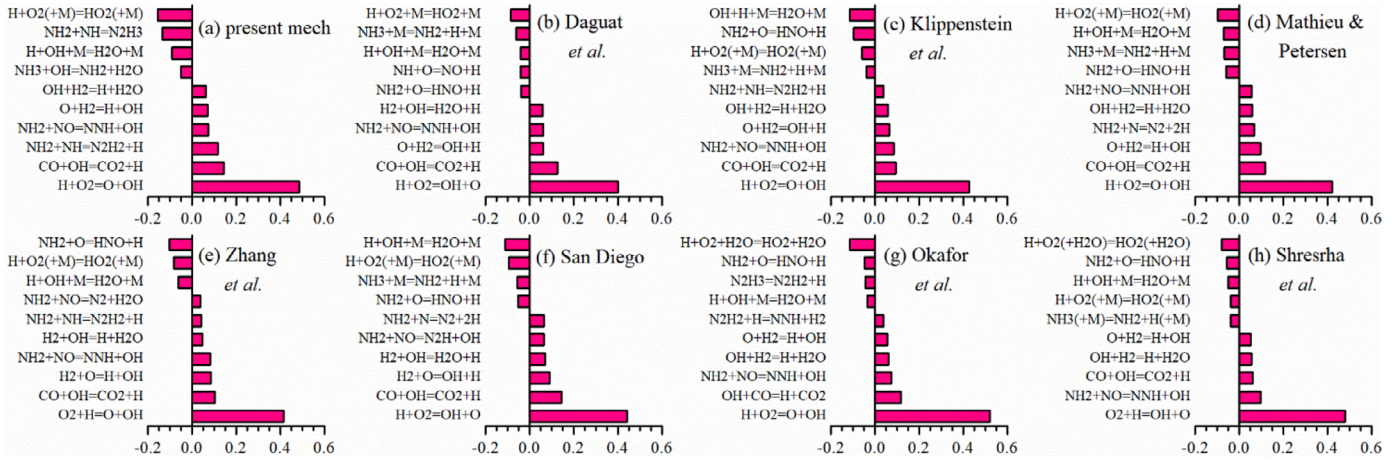


Fig. 14. Normalized sensitivity coefficients of the burning velocity of $\text{NH}_3/\text{SYN}_B/\text{air}$ flames for $x_{\text{SYN}} = 0.6$ and $\phi = 1$, calculated using the present and literature mechanisms.

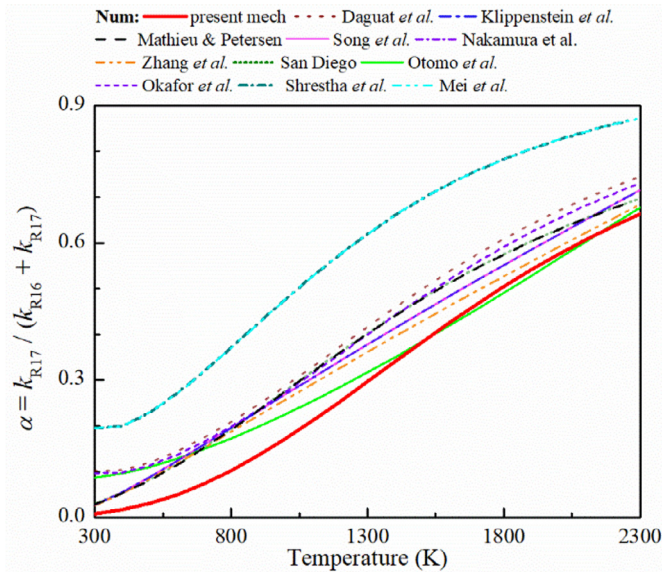


Fig. 15. The branching ratio $\alpha = k_{R17}/(k_{R17} + k_{R16})$ in different mechanisms.

twice as much as that used in present mechanism and Otomo et al. [32]. As notified by the '*' mark in Table 3, there are more than 50 important reactions with different rate parameters compared from different sources, and the cumulative effect can lead to considerable discrepancies. From this point of view, detailed analyses of the uncertainty of these reactions, which are not discussed in this work, could be helpful for future organization or optimization of NH_3 kinetic mechanisms.

4. Conclusion

The present study experimentally investigated laminar burning velocities of $\text{NH}_3/\text{syngas}/\text{air}$ flames at atmospheric pressure and 298 K using the heat flux method. Two types of syngas were used, i.e., SYN_A : 5 vol% H_2 + 95 vol% CO and SYN_B : 50 vol% H_2 + 50 vol% CO , and the measured conditions cover wide ranges of mixing ratios and equivalence ratios. It was found that the dependence of laminar burning velocity on mixing ratio and equivalence ratio of $\text{NH}_3/\text{SYN}_A/\text{air}$ flame is very alike that of $\text{NH}_3/\text{CO}/\text{air}$ flame; while that of $\text{NH}_3/\text{SYN}_B/\text{air}$ flame resembles the $\text{NH}_3/\text{H}_2/\text{air}$ flame more than the former two flames.

By the high-quality experimental results, several literature numerical mechanisms were tested, with no tested mechanism capable of predicting accurately all the experimental conditions. A new mechanism was proposed, and the results using this present mechanism agree well with experimental data of not only the burning velocities, but also the ignition delay times of NH_3 , NH_3/H_2 , NH_3/CO , $\text{NH}_3/\text{syngas}$ flames at various mixing ratios, equivalence ratios, and pressures. The present mechanism can also reproduce the trend of measured NO_x emission characteristics in literature. Detailed kinetic analyses using the present mechanism were carried out, showing the NH_3 oxidation processes in $\text{NH}_3/\text{syngas}/\text{air}$ flames and the most rate limiting reactions of predicting the laminar burning velocities. Important reactions with different rate parameters from different sources were labeled, which could be helpful for future organization or optimization of NH_3 kinetic mechanisms.

It should be noted that the present mechanism includes no interaction channels between C and N containing species, thus its good agreement with present experimental data prove again that this kind of interaction is insignificant for the burning velocity of NH_3 blended flames. On this premise and regarding the syngas oxidation reactions being the foundation of larger fuel molecule mechanisms, it is possible to incorporate the present mechanism for the kinetic modeling of NH_3 co-firing flames with larger fuel molecules, which is an interesting task for future investigation.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2019.11.032.

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